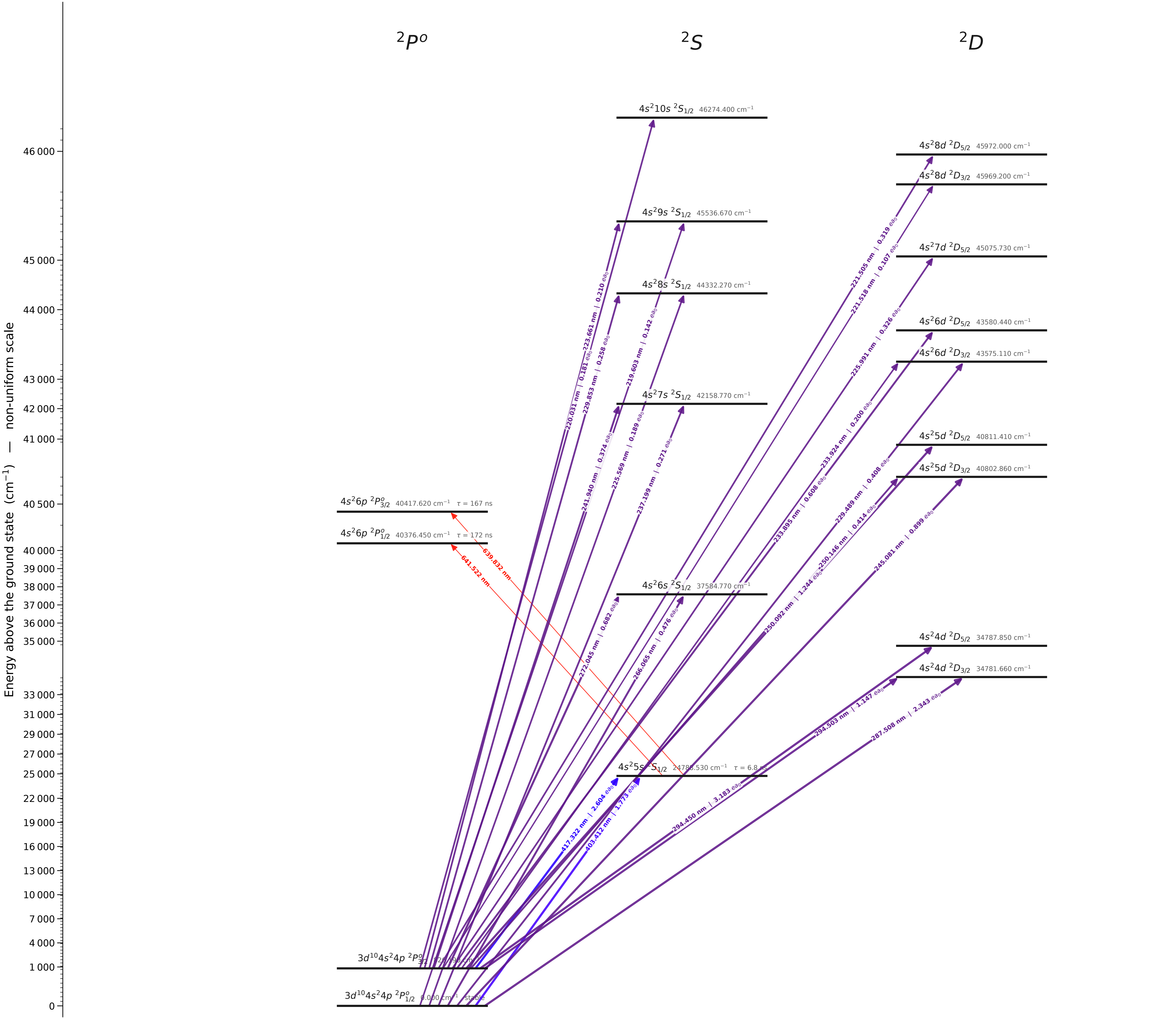


Gallium

Ga (Ga I). Strongest classified transitions below 2 μm (NIST): 19 levels, 25 lines.

Ga ⁺ ionisation limit 48387.63 cm^{−1} = 5.99930 eV (206.664 nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ^{−1})	Γ/2π up. (MHz)	branch (%)	Use
3d ¹⁰ 4s ² 4p ² P _{3/2} ^o – 4s ² 4d ² D _{5/2}	294.4497	294.3636	1018.1449	3.183	1.340e+08	–	–	294.45 nm laser-cooling
3d ¹⁰ 4s ² 4p ² P _{1/2} ^o – 4s ² 5s ² S _{1/2}	403.4124	403.2984	743.1414	1.773	4.850e+07	23.41	33	403.3 nm resonance line
3d ¹⁰ 4s ² 4p ² P _{3/2} ^o – 4s ² 5s ² S _{1/2}	417.3215	417.2039	718.3729	2.604	9.450e+07	23.41	64.3	417.2 nm line
4s ² 5s ² S _{1/2} – 4s ² 6p ² P _{3/2} ^o	639.8325	639.6557	468.5483	–	–	0.953	–	639.7 nm
4s ² 5s ² S _{1/2} – 4s ² 6p ² P _{1/2} ^o	641.5224	641.3451	467.3141	–	–	0.9253	–	641.3 nm

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

⁶⁹Ga hyperfine structure (I = 3/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
3d ¹⁰ 4s ² 4p ² P _{3/2} ^o	1338.9938(5)	–	F=1: -1673.74 F=2: +1004.25
3d ¹⁰ 4s ² 4p ² P _{3/2} ^o	191.5(9)	63.0(20)	F=0: -639.38 F=1: -510.88 F=2: -190.88 F=3: +446.62
4s ² 5s ² S _{1/2}	1069.5(15)	–	F=1: -1336.88 F=2: +802.12
4s ² 4d ² D _{3/2}	-36.3(22)	5(4)	F=0: +141.88 F=1: +100.97 F=2: +23.77 F=3: -80.53
4s ² 4d ² D _{5/2}	77.3(9)	5(4)	F=1: -402.12 F=2: -251.75 F=3: -22.24 F=4: +291.20
4s ² 6p ² P _{1/2} ^o	53.3(7)	–	F=1: -66.62 F=2: +39.98
4s ² 6p ² P _{3/2} ^o	22.40(20)	3.2(3)	F=0: -80.00 F=1: -60.80 F=2: -19.20 F=3: +51.20

Measured A, B constants; sources in the data file.

Arrow label: vacuum wavelength | reduced dipole matrix element |⟨J||er||J'⟩| in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where available)

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
3d ¹⁰ 4s ² 4p ² P _{3/2} ^o – 4s ² 4d ² D _{5/2}	294.450	71-69	115(7)
3d ¹⁰ 4s ² 4p ² P _{3/2} ^o – 4s ² 4d ² D _{3/2}	294.503	71-69	114(8)
3d ¹⁰ 4s ² 4p ² P _{1/2} ^o – 4s ² 5s ² S _{1/2}	403.412	69-71	33(4)
3d ¹⁰ 4s ² 4p ² P _{3/2} ^o – 4s ² 5s ² S _{1/2}	417.322	69-71	40(4)

Shift = ν(first isotope) – ν(second isotope).